

Loops and trace tables

Project: Python readiness portfolio

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Branching and boundary tests

Without opening the old lesson, explain one key idea from Branching and boundary tests. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Expressions and operator order

Without opening the old lesson, explain one key idea from Expressions and operator order. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Trace before running for Loops and trace tables. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Loops and trace tables. Proposed explanation: “The stop value is included”.
Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Count iterations and a stopping point

List every value visited and each running total. Then predict the result for range(2,2).

```
total = 0

for value in range(2, 6):
    total = total + value

print(total)
```

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. W1 Explain the range

List the values supplied by `range(2, 5)` and explain why 5 is absent. How many times does its body run?

My reasoning and evidence

6. W2 Trace before running

Copy the worked script. Change the range to `range(2, 5)`. Record turn, total before and total after each iteration. Predict the single output, then run and compare.

My reasoning and evidence

7. W3 Repair the reset

For the faulty script with `range(1, 4)`, record total after each addition. Explain the wrong final output and remove only the faulty reset.

My reasoning and evidence

8. W4 Build a repeated addition independently

A box starts with 5 counters. Each round adds 2 counters. Use `rounds = 3`, an initialized total and for turn in `range(rounds)`. Print the final total once after the loop. Trace the totals and predict fresh runs with rounds 0, 1 and 3. Assume rounds is a nonnegative integer.

My reasoning and evidence

9. W5 Check a contract limitation

In your counter script, `range(-2)` performs no iterations. Does returning the initial 5 make negative rounds

valid? Explain what the script checks and what it assumes.

My reasoning and evidence

10. Project checkpoint and reflection

Add a labelled loops and trace tables evidence sheet to Python readiness portfolio. Show how you can trace variable values before and after each loop iteration. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

11. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

12. Pictorial case practice: Loops and trace tables

Accumulate a short range. A running total adds the integers visited by `range(1,5)`.

Python 3: `total=0; for value in range(1,5): total+=value; print(total)`. The print is after the loop.

Running total after each visited value

	Current value	Running total
Visit 1	1	1
Visit 2	2	3
Visit 3	3	6
Visit 4	4	10

Read a cell using both its row and column labels.

Loops and trace tables

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Visit the range

Values are 1,2,3,4; the stop value 5 is excluded.



2. Trace running total

Totals after each addition are 1,3,6,10.



3. Print after the loop

The final displayed value is 10.

CHANGE ONE THING

Replace `range(1,5)` with `range(1,1)`.

Explain the changed result using the matching lesson.

List the four visited values and their running totals. Explain why 5 is absent and what prints for `range(1,1)`.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted outputs and the changed-case explanation. Use actual Python 3 results to check the prediction after reasoning first.

13. Project portfolio milestone

Project: Python readiness portfolio

Current milestone: Python readiness portfolio. Keep runnable code, predictions, actual outputs, one repaired error and an explanation of a function on changed inputs.

For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted outputs and the changed-case explanation. Use actual Python 3 results to check the prediction after reasoning first.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

14. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Accumulate a short range. A running total adds the integers visited by `range(1,5)`.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? The final displayed value is 10.

1. Trace running total
2. Print after the loop
3. Visit the range

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Print after the loop The final displayed value is 10. A loop repeats an operation for each visited value. A trace table distinguishes the current value from the accumulator; the range stop is not included.

Practice 2: explain the deciding evidence

Changed-case practice: Replace `range(1,5)` with `range(1,1)`. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed input or the procedure.
2. No values are visited and the final output is 0.
3. This one example guarantees correct results for every future input.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: No values are visited and the final output is 0. The accumulator starts at zero and an empty loop never changes it.

Repair a missing building block

Review Branching and boundary tests (python-bridge/5). Goal: Trace the selected branch in an if elif else chain. Short practice: Trace -2, 4 and 5 through the branches. Explain why the negative check comes first and what label -2 gets if it is removed.. Return to this week and explain what you changed.

Review Expressions and operator order (python-bridge/3). Goal: Evaluate numeric expressions using Python's grouping rules. Short practice: Trace the operation order in both expressions. Identify which picture step changes with parentheses and predict both results when the multiplier becomes four.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
